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Assignment-9

[Java/Experiment-9 at main · diyaa-13/Java]

JdbcApplication.java U ● mysql-connector-j-9.6.0.jar

Experiment-9 > JdbcApplication.java > JdbcApplication

```java
import java.sql.*;
import java.util.Scanner;

public class JdbcApplication {
    static final String URL  = "jdbc:mysql://localhost:3306/college_db";
    static final String USER = "root";
    static final String PASS = "stargirl15";

    //Get connection
    static Connection getConnection() {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            return DriverManager.getConnection(URL, USER, PASS);
        }
        catch (Exception e) {
            throw new RuntimeException(e);
        }
    }

    //Create table
    static void createTable() {
        String sql = "Create table if not exists students ("
                   + "roll_no INT PRIMARY KEY, "
                   + "name VARCHAR(50), "
                   + "branch VARCHAR(30), "
                   + "marks DOUBLE)";
        try (Connection con = getConnection(); Statement st = con.createStatement()) {
            st.executeUpdate(sql);
            System.out.println("  Table ready.");
        }
        catch (SQLException e) {
            System.out.println("  [ERROR] " + e.getMessage());
        }
    }
```

Experiment-9 > JdbcApplication.java > JdbcApplication

```java
public class JdbcApplication {

    //Insert record
    static void insertRecord(int roll, String name, String branch, double marks) {
        String sql = "INSERT INTO students VALUES (?, ?, ?, ?)";
        try (Connection con = getConnection(); PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setInt(1, roll);
            ps.setString(2, name);
            ps.setString(3, branch);
            ps.setDouble(4, marks);
            ps.executeUpdate();
            System.out.println("Record inserted.");
        }
        catch (SQLException e) {
            System.out.println("  [ERROR] " + e.getMessage());
        }
    }

    //Fetch all records
    static void fetchRecords() {
        String sql = "SELECT * FROM students ORDER BY roll_no";
        try (Connection con = getConnection();
            Statement st   = con.createStatement();
            ResultSet rs   = st.executeQuery(sql)) {
                System.out.println("\n  +------+--------------------+-----------+--------+");
                System.out.println("  | Roll | Name               | Branch    | Marks  |");
                System.out.println("  +------+--------------------+-----------+--------+");
                boolean any = false;
                while (rs.next()) {
                    System.out.printf("  | %-4d | %-19s | %-9s | %-6.1f |%n",
                    rs.getInt("roll_no"), rs.getString("name"),
                    rs.getString("branch"), rs.getDouble("marks"));
                    any = true;
            }
            if (!any) System.out.println("  No records found.");
            System.out.println("  +------+--------------------+-----------+--------+");
        }
        catch (SQLException e) {
            System.out.println("  [ERROR] " + e.getMessage());
```

Experiment-9 > JdbcApplication.java > JdbcApplication

```java
public class JdbcApplication {

    //Update record
    static void updateRecord(int roll, double newMarks) {
        String sql = "UPDATE students SET marks=? WHERE roll_no=?";
        try (Connection con = getConnection(); PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setDouble(1, newMarks);
            ps.setInt(2, roll);
            int rows = ps.executeUpdate();
            System.out.println(rows > 0 ? "  Record updated." : "  Roll No not found.");
        }
        catch (SQLException e) {
            System.out.println("  [ERROR] " + e.getMessage());
        }
    }

    //Delete record
    static void deleteRecord(int roll) {
        String sql = "DELETE FROM students WHERE roll_no=?";
        try (Connection con = getConnection(); PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setInt(1, roll);
            int rows = ps.executeUpdate();
            System.out.println(rows > 0 ? "  Record deleted." : "  Roll No not found.");
        }
        catch (SQLException e) {
            System.out.println("  [ERROR] " + e.getMessage());
        }
    }

//Main
Run | Debug
public static void main(String[] args) {
    createTable();
     Scanner sc = new Scanner(System.in);
    System.out.println("JDBC Student Database");
```

Experiment-9 > JdbcApplication.java > JdbcApplication

```java
public class JdbcApplication {
public static void main(String[] args) {
    while (true) {
        System.out.println("\n1.Insert  2.View  3.Update  4.Delete  0.Exit");
        System.out.print("Choice: ");
        int ch;
        try {
            ch = Integer.parseInt(sc.nextLine().trim());
        }
        catch (NumberFormatException e) {
            System.out.println("Invalid!"); continue;
        }

        try {
            switch (ch) {
                case 1:
                    System.out.print("Roll: ");
                    int r  = Integer.parseInt(sc.nextLine().trim());
                    System.out.print("Name: ");
                    String n = sc.nextLine().trim();
                    System.out.print("Branch: ");
                    String br = sc.nextLine().trim();
                    System.out.print("Marks: ");
                    double m = Double.parseDouble(sc.nextLine().trim());
                    insertRecord(r, n, br, m);
                    break;

                case 2:
                    fetchRecords();
                    break;

                case 3:
                    System.out.print("Roll to update: ");
                    int ur = Integer.parseInt(sc.nextLine().trim());
                    System.out.print("New Marks: ");
                    double um = Double.parseDouble(sc.nextLine().trim());
                    updateRecord(ur, um);
                    break;
```

```java
                case 4:
                    System.out.print("Roll to delete: ");
                    deleteRecord(Integer.parseInt(sc.nextLine().trim()));
                    break;

                case 0:
                    System.out.println("Goodbye!");
                    sc.close();
                    return;

                default:
                    System.out.println("Invalid option.");
            }
        }
        catch (NumberFormatException e) {
            System.out.println("  [ERROR] Invalid numeric input.");
        }
    }
}
}
```

Output:

```
PS C:\Users\vasum\Desktop\Diya\Java\Java\Experiment-9> java -
  Table ready.
JDBC Student Database

1.Insert  2.View  3.Update  4.Delete  0.Exit
Choice: 1
Roll: 211
Name: Diya
Branch: AIML
Marks: 30
Record inserted.

1.Insert  2.View  3.Update  4.Delete  0.Exit
Choice: 1
Roll: 207
Name: Wani
Branch: AIML
Marks: 35
Record inserted.

1.Insert  2.View  3.Update  4.Delete  0.Exit
Choice: 2

  +------+---------------------+-----------+--------+
  | Roll | Name                | Branch    | Marks  |
  +------+---------------------+-----------+--------+
  | 207  | Wani                | AIML      | 35.0   |
  | 211  | Diya                | AIML      | 30.0   |
  +------+---------------------+-----------+--------+

1.Insert  2.View  3.Update  4.Delete  0.Exit
Choice: 3
Roll to update: 211
New Marks: 34
  Record updated.

1.Insert  2.View  3.Update  4.Delete  0.Exit
Choice: 4
Roll to delete: 207
  Record deleted.
```

```
1.Insert  2.View  3.Update  4.Delete  0.Exit
Choice: 2

  +------+---------------------+-----------+--------+
  | Roll | Name                | Branch    | Marks  |
  +------+---------------------+-----------+--------+
  | 211  | Diya                | AIML      | 34.0   |
  +------+---------------------+-----------+--------+

1.Insert  2.View  3.Update  4.Delete  0.Exit
Choice: 5
Invalid option.

1.Insert  2.View  3.Update  4.Delete  0.Exit
Choice: 0
Goodbye!
PS C:\Users\vasum\Desktop\Diya\Java\Java\Experiment-9>
```